Section 5.3 — Rational Roots Theorem — Answer Key

Math 103: College Algebra

Multiplying Quickly

Find the leading term of each product below.

1. $2x^4$
2. $8x^6$
3. $12x^9$

Find the constant term of each product below.

4. 28
5. 28
6. 96

Possible Rational Roots

List all possible rational roots of each polynomial below.

7. $\pm 1, \pm 2, \pm 3, \pm 6$
8. $\pm 1, \pm 5, \pm \frac{1}{3}, \pm \frac{5}{3}$
9. $\pm 1, \pm 2, \pm \frac{1}{2}, \pm \frac{1}{3}, \pm \frac{2}{3}, \pm \frac{1}{6}$
10. $\pm 1, \pm 2, \pm 3, \pm 5, \pm 6, \pm 10, \pm 15, \pm 30, \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{5}{2}, \pm \frac{15}{2}, \pm \frac{1}{3}, \pm \frac{2}{3}, \pm \frac{5}{3}, \pm \frac{10}{3}, \pm \frac{1}{4}, \pm \frac{3}{4}, \pm \frac{5}{4}, \pm \frac{15}{4}, \pm \frac{1}{6}, \pm \frac{5}{6}, \pm \frac{1}{12}, \pm \frac{5}{12}$

Descartes's Rule of Signs

Use Descartes's Rule of Signs to describe the possible numbers of positive and negative roots of each polynomial.

11. Positive: 2 or 0; Negative: 2 or 0
12. Positive: 1; Negative: 2 or 0
13. Positive: 0; Negative: 1
14. Positive: 5, 3, or 1; Negative: 0

Trial and Error

For the polynomial $12x^4 - 23x^3 - 58x^2 + 37x + 60$, test each of the following possible rational roots, and state which (if any) of the following are true:

a. It is a root

b. It gives an upper bound on the roots

c. It gives a lower bound on the roots

d. None of these

15. None of these:

1	12	−23	−58	37	60
		12	−11	−69	−32
	12	−11	−69	−32	28

16. Lower bound:

−2	12	−23	−58	37	60
		−24	94	−72	70
	12	−47	36	−35	130

17. Root:

3	12	−23	−58	37	60
		36	39	−57	−60
	12	13	−19	−20	0

18. Upper bound:

5	12	−23	−58	37	60
		60	185	635	3360
	12	37	127	672	3420

19. Root:

$-\frac{4}{3}$	12	−23	−58	37	60
		−16	52	8	−60
	12	−39	−6	45	0

Factor Fully

Each polynomial below can be factored as a product of linear factors with integer coefficients. Factor them. Begin by using Descartes's Rule of Signs to analyze the signs of the roots.

20. $(x+1)(x+3)(x-2)$

21. $(x+2)(x+5)(2x-3)$

22. $(x+1)(2x+1)(2x+5)(3x-2)$

23. $(x+1)(x+2)(x+3)(x-4)(x-5)$